ERG Analysis API v2.4.0

Tier 3 — T3-E: Regulatory Readiness Summary

Document ID	T3-E_Regulatory_Readiness_Summary_v1_0
Product	ERG Analysis API v2.4.0
Organisation	AI Fairness CIO · Registered Charity No. 1218464 · London, United Kingdom
Pipeline version	v2.4.0 (PIPELINE_VERSION = "2.4.0")
Repo	github.com/AI-Fairness-com/ERG-Analysis-API
OSF pre-reg	https://doi.org/10.17605/OSF.IO/6WA42
Regulatory frame	MHRA AI/ML SaMD Guidance (2021); UKCA; ISO 13485:2016; ISO 14971:2019
Tier 3 documents	T3-A ISCEV Checklist · T3-B Traceability Matrix · T3-C Bias Audit · T3-D Intended Use · T3-E (this document)
Author	Hamid Tavakoli, Founder & CEO, AI Fairness CIO
Date	24 June 2026
Status	Signed-off by Hamid Tavakoli

Executive Summary

Regulatory Readiness Verdict — ERG Analysis API v2.4.0

ERG Analysis API v2.4.0 is READY FOR TIER 4 (external clinical validation).

The pipeline has completed Tiers 1–3 of the Production Readiness SOP. All hard technical requirements for external validation readiness are met: ISCEV compliance is documented and enforced at runtime; normative data traceability is complete; bias risks are identified and mitigated within the scope of Baker et al. (2025); an intended use statement and contraindications are formally documented; and all outputs carry mandatory clinical decision-support disclaimers.

Three gaps must be closed before clinical deployment (Tier 5): (1) ML classifiers are not yet trained — the pipeline currently provides Z-score decision support only, not disease classification; (2) no external clinical validation has been conducted; (3) no post-market surveillance plan exists. These are Tier 4 activities and do not prevent Tier 4 commencement.

Two structural gaps (sex/ethnicity stratification; ISO 13485 QMS) are known limitations of the Baker 2025 normative dataset and the current organisational stage respectively. Both are documented, disclosed, and consistent with the pre-registered scope.

1. Production Readiness Tier Status

The ERG Analysis API is governed by a five-tier Production Readiness SOP. The table below records the completion status of each tier as of 24 June 2026.

Tier	Name	Content	Status
Tier 1	**Synthetic Validation**	41/41 PASS. Generator defects G1–G7 and A1/S03 resolved. T1_Synthetic_Validation_Report_v2_4_0_COMPLETE.docx produced. OSF Wiki updated.	**COMPLETE**
Tier 2	**Code Hardening**	pytest suite 23/23 PASS (14 pipeline + 9 regression tests). Normative JSON externalised. DTL Deming regression implemented. Signal orientation detector (DA only). ISCEV 350 ms re-validation 30/30 PASS. Documented in T2_Code_Hardening_Report_v1_0.	**COMPLETE**
Tier 3	**Clinical & Regulatory**	T3-A ISCEV Compliance Checklist ✓ T3-B Normative Traceability Matrix ✓ T3-C Bias & Fairness Audit ✓ T3-D Intended Use Statement ✓ T3-E Regulatory Readiness Summary (this document)	**IN PROGRESS**
Tier 4	**External Validation**	Prospective validation on real patient data from external clinical sites. ML classifier training and AUC evidence. Post-market surveillance plan. Regulatory submission package.	**NOT STARTED**

Tier	Name	Content	Status
Tier 5	**Clinical Deployment**	UKCA/CE marking (if applicable). Ethics approval. Clinical governance sign-off. Live deployment with post-market surveillance.	**NOT STARTED**

2. Regulatory Requirements — RAG Status

The table below maps each regulatory and quality requirement to the evidence produced across Tiers 1–3, and assigns a status: MET (evidence complete), PARTIAL (evidence exists but incomplete), PENDING (planned for Tier 4), or GAP (structural limitation with no current path to closure within v2.4.0 scope).

Status key:

- MET — requirement is fully satisfied by documented evidence in the Tier 1–3 package.
- PARTIAL — requirement is partially satisfied; a gap is identified with a path to closure.
- PENDING — requirement is a Tier 4 activity; not blocking Tier 4 commencement.
- GAP — structural limitation; documented, disclosed, and consistent with pre-registered scope.

Domain	Requirement	Evidence Summary	T3 Ref	Status
Intended Use & Indication	Documented intended use, patient population, contraindications, and clinical setting	T3-D_Intended_Use_Statement_v1_0 — 7 sections covering intended use statement, patient population, contraindications (C1–C5), limitations (L1–L9), output interpretation guide, mandatory disclaimers	**T3-D**	**MET**
Software Purpose & Classification	Classification as SaMD decision-support tool; not standalone diagnostic	T3-D §5.1: classified as decision-support software. Both report disclaimers explicitly state output does not constitute a medical diagnosis. Layer 2 recommended_action field enforces clinician review.	**T3-D**	**MET**
Clinical Safety — Disclaimer	Mandatory disclaimer on every clinical output	Disclaimer text reproduced verbatim in T3-D §6. Present in every PDF report (gold-bordered block, erg_report_generator.py) and in Layer 2 JSON (disclaimer field, erg_v2_4_0.py line ~1511).	**T3-D**	**MET**
Standards Compliance — ISCEV 2022	Demonstration of compliance with recognised clinical standard for ERG	T3-A_ISCEV_Compliance_Checklist_v1_0: 14 PASS, 3 AMBER (all remediated), 1 FAIL (G1 version string — remediated). Runtime ISCEV compliance validation with Layer 4 audit flag.	**T3-A**	**MET**

Domain	Requirement	Evidence Summary	T3 Ref	Status
Normative Data Traceability	Full audit trail from published normative data to pipeline output	T3-B_Normative_Traceability_Matrix_v1_0: 96 non-null parameter cells cross-walked from Baker et al. (2025) JSON → pipeline variable → report field. 5 PASS, 3 AMBER (all remediated).	**T3-B**	**MET**
Bias & Fairness	Documented assessment of systematic bias risks; stratified reference norms	T3-C_Bias_Fairness_Audit_v1_0: Baker 2025 three-stratum age framework confirmed; DTL Deming correction assessed; traffic light symmetry checked. 5 PASS, 2 AMBER (B4 remediated in code; B7 documented in T3-D).	**T3-C**	**MET**
Pre-registration & Transparency	Publicly registered study protocol and statistical analysis plan before data collection	OSF pre-registration DOI: 10.17605/OSF.IO/6WA42. Includes locked parameters, aspirational AUC targets, classifier architecture, normative reference. T3-D §4 cross-walk distinguishes implemented vs aspirational items.	**T3-D**	**MET**
Version Control	Unambiguous version identification across all pipeline artefacts	PIPELINE_VERSION = "2.4.0" in ERGConfig (line 111). Written to Layer 4 audit JSON, FHIR Observation meta, and PDF subtitle. Version comment header corrected (G1 T3-A remediation).	**T3-A**	**MET**
Signal Quality Grading	Documented objective signal quality assessment with clinician guidance	ERGAudit.run_full_audit() produces quality grade A–D based on SNR, OP availability, bandwidth, spike count, saturation, and ISCEV compliance. Grade and description rendered in PDF Layer 3 panel. Clinician guidance in T3-D §1.4.	**T3-A**	**MET**
FHIR R4 Interoperability	Structured clinical output in recognised health informatics standard	ERGFHIRGenerator produces HL7 FHIR R4 Observation resource: LOINC 70943-7, SNOMED CT interpretation codes, versioned CodeSystem URI (E5 T3-A remediation), component codes for all ERG parameters.	**T3-A**	**MET**
Audit Trail — Layer 4	Complete technical audit log per recording for post-market surveillance	Layer 4 technical audit dict records: report_id, timestamp, pipeline_version, electrode_type, age_stratum, protocol, z_scores, filter_log, processing_time_ms, inverted_polarity_detected, DTL correction flag, LA 3 r^2 flag, flash duration compliance, paediatric flag.	**T3-A/B**	**MET**
ML Classifier — Stage 1	Trained binary Normal / Abnormal classifier; AUC ≥0.94 (pre-registered aspirational target)	NOT YET IMPLEMENTED. Pre-registered as aspirational. v2.4.0 uses Z-score traffic light in lieu of trained classifier. Layer 3 Specialist output is labelled PENDING. No AUC evidence available.	**T3-D**	**PENDING**

Domain	Requirement	Evidence Summary	T3 Ref	Status
ML Classifier — Stage 2	Trained six-class disease classifier; macro-AUC ≥0.80 (pre-registered aspirational target)	NOT YET IMPLEMENTED. Pre-registered as aspirational. Disease classification is deferred to Tier 4 / external validation phase. Stage 2 output not produced in v2.4.0.	**T3-D**	**PENDING**
Clinical Validation — External	Prospective validation on real patient data from external clinical sites (Tier 4)	NOT YET STARTED. Tier 1 (synthetic validation 41/41 PASS) and Tier 2 (code hardening) are complete. Tier 3 (this document set) is in progress. Tier 4 external validation is the next stage.	**T4**	**PENDING**
Sex-stratified Normative Data	Reference ranges stratified by sex to detect sex-related bias	NOT AVAILABLE in Baker et al. (2025). No sex field is recorded by the pipeline. Documented as known limitation L1 in T3-D. To be addressed if a future normative dataset provides sex-stratified norms.	**T3-C**	**GAP**
Paediatric Normative Data	Reference ranges for patients under 18 years	NOT AVAILABLE in Baker et al. (2025). Pipeline emits visible paediatric disclaimer block in PDF report (D5 T3-A remediation) and WARNING in Layer 4 audit. Documented as contraindication C1 in T3-D.	**T3-D**	**GAP**
Post-market Surveillance Plan	Plan for ongoing monitoring of real-world performance after clinical deployment	NOT YET PRODUCED. Required before clinical deployment. To be produced as part of Tier 4 / regulatory submission package.	**T4**	**PENDING**
ISO 13485 Quality Management	Quality management system covering design controls, risk management, change control	PARTIAL. AI Fairness CIO is a registered charity operating under Charity Commission governance. A formal ISO 13485 QMS has not been established for v2.4.0. Production Readiness SOP v1.0 provides the five-tier roadmap as a proxy design control document.	**T4**	**PARTIAL**

3. Outstanding Gaps and Path to Closure

Five gaps are identified. Gaps G1–G4 are Tier 4 activities that must be completed before clinical deployment. Gap G5 is an organisational readiness item that should be initiated in parallel with Tier 4.

#	Gap	Description	Path to closure
G1	**ML classifiers not trained**	Stage 1 (binary) and Stage 2 (six-class disease) classifiers are pre-registered but not yet implemented. v2.4.0 uses Z-score traffic light only.	Tier 4 activity. Requires labelled clinical training dataset, model training, cross-validation, and AUC evidence against pre-registered targets. Layer 3 Specialist output will be populated when classifiers are deployed.
G2	**External clinical validation absent**	All validation to date is synthetic (Tier 1, 41/41 PASS) or code-level (Tier 2). No prospective validation on real patient data from external clinical sites has been conducted.	Tier 4 activity. Required before any clinical deployment. Must include at least one external electrophysiology department, real patient recordings across all five protocols and both supported electrode types, and comparison against consensus clinical expert ground truth.
G3	**Sex and ethnicity stratification absent**	Baker et al. (2025) does not provide sex- or ethnicity-stratified normative data. These demographic variables are not recorded or used by the pipeline.	Monitor literature for future normative datasets with demographic stratification. If deployed in a population with known demographic characteristics, document this as a residual bias risk in the post-market surveillance plan.
G4	**Post-market surveillance plan absent**	No formal plan exists for monitoring real-world performance, adverse event reporting, or periodic safety updates after clinical deployment.	Required before clinical deployment. Must include: key performance indicators (e.g. rate of Grade D recordings, AMBER/RED override rate, clinician feedback), adverse event reporting pathway, and periodic review schedule.
G5	**ISO 13485 QMS not established**	A formal quality management system has not been established. The Production Readiness SOP v1.0 provides a roadmap but is not equivalent to a certified QMS.	Required before regulatory submission. Minimum viable QMS for a small charity SaMD developer: documented design controls, risk management file (ISO 14971), change control log, and supplier/component qualification records.

4. Tier 3 Document Inventory

The following documents constitute the complete Tier 3 Clinical & Regulatory package for ERG Analysis API v2.4.0. All are produced against the patched pipeline (erg_v2_4_0.py post T3-A/B/C remediation patches) and the normative JSON (normative_data_baker2025.json schema_version 1.0).

Ref	Document	Filename	Version	Status
T1	Synthetic Validation Report	T1_Synthetic_Validation_Report_v2_4_0_COMPLETE.docx	v1.0	Signed off

Ref	Document	Filename	Version	Status
T2	Code Hardening Report	T2_Code_Hardening_Report_v1_0.docx	v1.0	DRAFT — sign-off pending
T3-A	ISCEV Compliance Checklist	*T3-A_ISCEV_Compliance_Checklist_v1_0.docx*	v1.0	sign-off pending
T3-B	Normative Reference Traceability Matrix	*T3-B_Normative_Traceability_Matrix_v1_0.docx*	v1.0	DRAFT — sign-off pending
T3-C	Bias & Fairness Audit	*T3-C_Bias_Fairness_Audit_v1_0.docx*	v1.0	DRAFT — sign-off pending
T3-D	Intended Use & Limitations Statement	*T3-D_Intended_Use_Statement_v1_0.docx*	v1.0	DRAFT — sign-off pending
T3-E	Regulatory Readiness Summary (this document)	*T3-E_Regulatory_Readiness_Summary_v1_0.docx*	v1.0	DRAFT — sign-off pending

Pipeline files delivered with this Tier 3 package:

File	Changes in Tier 3	Patches applied
erg_v2_4_0.py	G1 version comment fixed; A6 flash duration validation; E5 FHIR CodeSystem URI versioned; F1 schema assertion; F6 b/a ratio loaded from JSON; F4 LA3 DTL r^2 flag; B4 LA 30 Hz absolute floor	T3-A: G1, A6, E5 T3-B: F1, F6, F4 T3-C: B4
erg_report_generator.py	D5 paediatric disclaimer block added (PAEDIATRIC_AGE_GROUPS, PAEDIATRIC_DISCLAIMER_TEXT, build_paediatric_disclaimer_block)	T3-A: D5

5. Recommended Next Steps — Tier 4 Initiation

The following actions are recommended to initiate Tier 4 external validation. They are ordered by dependency, not priority.

#	Action	Description	Owner / Dependency
1	**Sign off Tier 3**	Hamid Tavakoli counter-signs all five T3 documents (T3-A through T3-E) to reclassify them from DRAFT to APPROVED. Update CHANGELOG.md with Tier 3 completion.	Hamid Tavakoli Immediate
2	**Commit Tier 3 artefacts**	Commit the five T3 documents and both patched pipeline files to the GitHub repository under a /docs/tier3/ directory. Tag the commit as v2.4.0-tier3-complete.	Hamid Tavakoli After sign-off
3	**Identify external validation site(s)**	Identify at least one external clinical electrophysiology department willing to provide real patient ERG recordings across all five ISCEV protocols and both supported electrode types. Agree data sharing agreement and anonymisation protocol.	Hamid Tavakoli Clinical network
4	**Ethics application**	Submit ethics application for prospective collection of real patient ERG data for algorithm validation. Reference OSF pre-registration (DOI: 10.17605/OSF.IO/6WA42) in application.	Hamid Tavakoli After site identified
5	**Initiate QMS documentation**	Begin ISO 13485-aligned quality management documentation: design history file, risk management file (ISO 14971), change control log. Minimum viable QMS for a small charity SaMD developer.	Hamid Tavakoli In parallel
6	**Draft post-market surveillance plan**	Define KPIs (Grade D rate, AMBER/RED override rate, clinician feedback), adverse event reporting pathway, and periodic review schedule. Required before Tier 5 clinical deployment.	Hamid Tavakoli Tier 4 phase

6. Sign-off

This document and the complete Tier 3 package will be reclassified APPROVED upon counter-signature by Hamid Tavakoli. Sign-off of T3-E constitutes sign-off of the entire Tier 3 Clinical & Regulatory package.

Role	Name	Date
T3-E Author (HT)	AI Fairness CIO ERG Pipeline	24 June 2026
Founder & CEO (Tier 3 Approval)	Hamid Tavakoli	27 June 2026

End of T3-E Regulatory Readiness Summary v1.0 — AI Fairness CIO (Reg. No. 1218464)